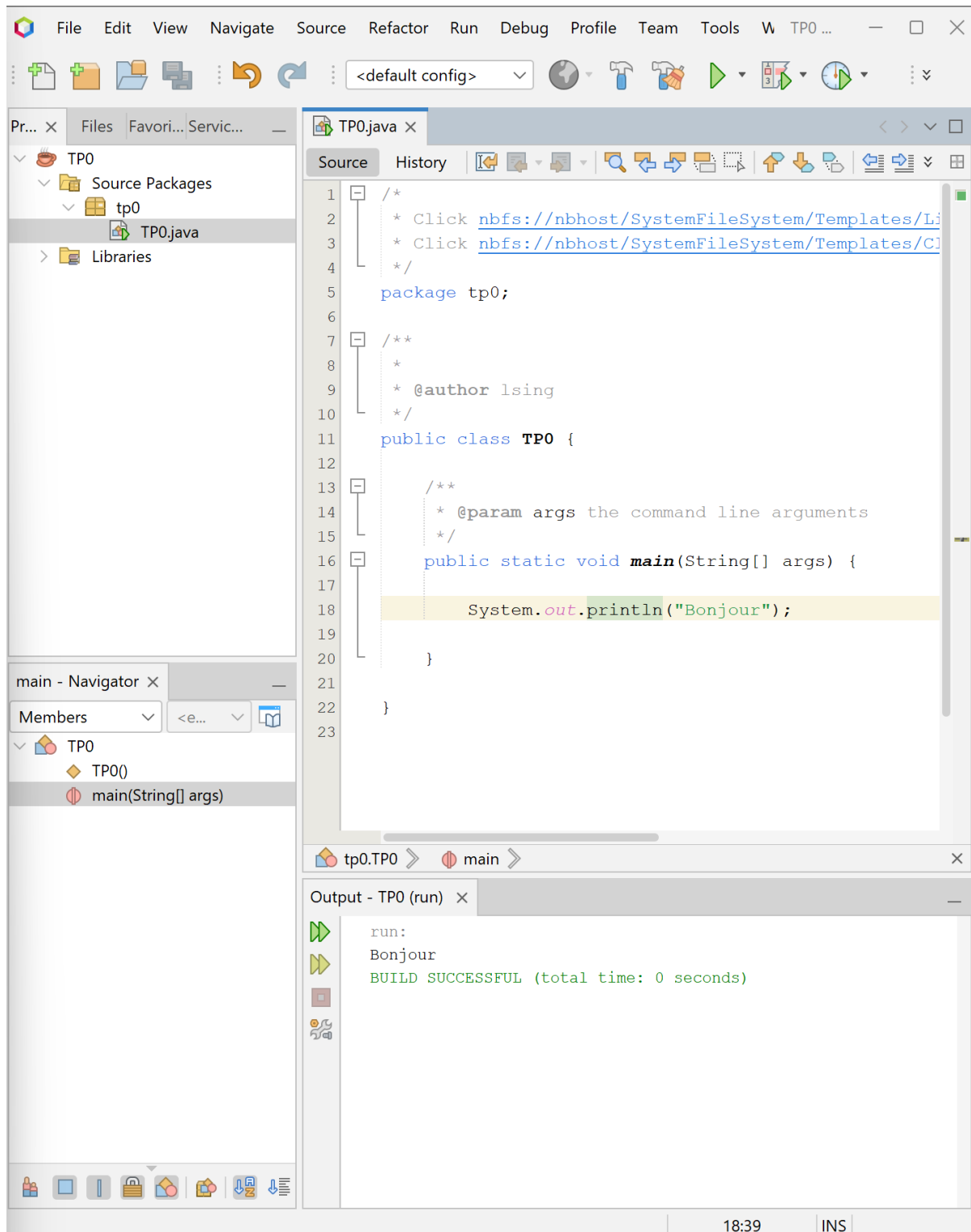


Partie 1 :

Here TP0 instead of Exo1

1 + 2 . Here im showing the output and the creation of the first project



Showing the output

The screenshot displays the NetBeans IDE interface. The main editor window shows the source code for `TP0.java`. The code is as follows:

```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/
4   */
5  package tp0;
6
7  /**
8   *
9   * @author lsing
10  */
11  public class TP0 {
12
13      /**
14       * @param args the command line arguments
15       */
16      public static void main(String[] args) {
17
18          System.out.println("Bonjour");
19
20      }
21
22  }
23
```

The `main` method is highlighted in yellow. The `Output - TP0 (jar)` window at the bottom shows the following output:

```
Compiling 1 source file to C:\Users\lsing\OneDrive - Efrei\Doc
compile:
Created dir: C:\Users\lsing\OneDrive - Efrei\Documents\NetBean
Copying 1 file to C:\Users\lsing\OneDrive - Efrei\Documents\Ne
Nothing to copy.
Building jar: C:\Users\lsing\OneDrive - Efrei\Documents\NetBea
To run this application from the command line without Ant, try
java -jar "C:\Users\lsing\OneDrive - Efrei\Documents\NetBeansP
deploy:
jar:
BUILD SUCCESSFUL (total time: 1 second)
```

The status bar at the bottom indicates the time is 18:39 and the window title is INS Windows (CRLF).

Changing the main code.

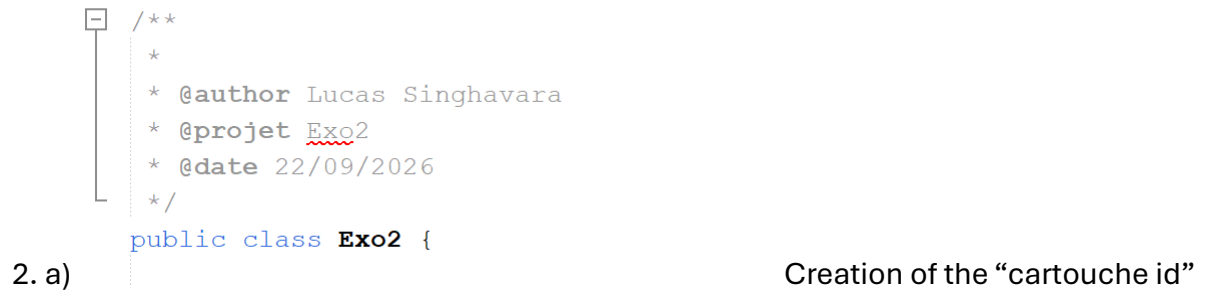
The screenshot displays an IDE interface with the following components:

- Top Menu Bar:** File, Edit, View, Navigate, Source, Refactor, Run, Debug, Profile, Team, Tools, W, TP0 ...
- Left Panel (Project Explorer):** Shows the project structure with 'TP0' as the root, containing 'Source Packages' (tp0) and 'Test Packages' (Test Packages, Libraries, Test Libraries). The file 'TP0.java' is selected under 'tp0'.
- Bottom Left Panel (main - Navigator):** Shows the 'Members' view for 'TP0', listing 'TP0()' and 'main(String[] args)'.
- Source Editor:** Displays the code for 'TP0.java'. The code includes a package declaration, an import for 'Scanner', a Javadoc comment for the author 'lsing', and a 'main' method that prompts for a name and prints a greeting. The variable 'prenom' is highlighted in green.
- Bottom Panel (Output - TP0 (run)):** Shows the execution output, which includes the prompt 'run:', the user input 'Bonsoir, comment allez-vous ?', the response 'Très bien', and a 'BUILD SUCCESSFUL' message with a total time of 9 seconds.

```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Li
3  * Click nbfs://nbhost/SystemFileSystem/Templates/C
4  */
5  package tp0;
6
7  import java.util.Scanner;
8
9  /**
10   *
11   * @author lsing
12   */
13  public class TP0 {
14
15      /**
16       * @param args the command line arguments
17       */
18      public static void main(String[] args) {
19
20          String prenom;
21          Scanner sc;
22          sc = new Scanner(System.in);
23          System.out.println("Bonsoir, comment allez-vous ?");
24          prenom = sc.nextLine();
25      }
26  }
```

run:
Bonsoir, comment allez-vous ?
Très bien
BUILD SUCCESSFUL (total time: 9 seconds)

Partie 2 :



```
14
15
16 /**
17  * @param args the command line arguments
18  */
19 public static void main(String[] args) {
20     // TODO code application logic here
21     //Declaration des variables
22     int nb; // nombre d'entier a additionner
23     int result; // resultat
24     int ind; //indice
25     nb=5;
26     result=0;
27
28     // Addition des nb premiers entiers
29     ind=1;
30     while (ind <= nb) {
31         result=result+ind;
32     }
33
34     // Affichage du resultat
35     System.out.println();
36     System.out.println("La somme des " + nb + "er");
37 }
38
39
```

exo2.Exo2 > main >

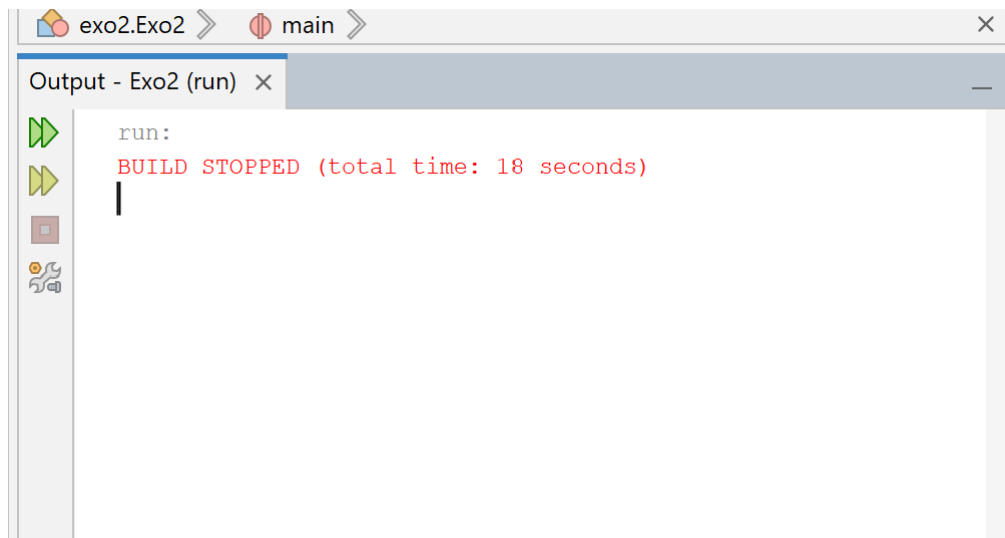
b)

the code in the main

Copying

Split	>
Code Folds	>
IDE Log	
Toolbars	>
✓ Show Editor Toolbar	
✓ Show Line Numbers	
Show Non-printable Characters	
✓ Show Breadcrumbs	
✓ Show Indent Guide Lines	
Show Inline Hints	Alt+I
✓ Show Inline Values	
✓ Show Diff Sidebar	
✓ Show Versioning Labels	
Synchronize Editor with Views	
Show Only Editor	Ctrl+Shift+Enter
Full Screen	Alt+Shift+Enter

3. Showing the lines



4. Example of the failed output

5. a) -> c) Testing debug

The screenshot shows an IDE with the following components:

- Top Bar:** File, Edit, View, Navigate, Source, Refactor, Run, Debug, Profile, Team, Tools, W, Exo2...
- Left Panel:**
 - Project Explorer: 'Common-Cleaner' runnir, 'main' suspended at 'Exo2', **Exo2.main:28**, 'Notification Thread' runnir.
 - Evaluate Expression: Empty.
 - main - Naviga... Breakpoints: Empty.
- Source Editor:** Shows the code for `Exo2.java`. The current line is 28, `ind=1;`. The code is as follows:

```
14
15 /**
16  * @param args the command line arguments
17  */
18 public static void main(String[] args) { args = #
19     // TODO code application logic here
20     //Declaration des variables
21     int nb; // nombre d'entier a additionner nb
22     int result; // resultat result = 0
23     int ind; //indice
24     nb=5; nb = 5
25     result=0; result = 0
26
27     // Addition des nb premiers entiers
28     ind=1;
29     while (ind <= nb) {
30         result=result+ind;
31     }
32
33     // Affichage du resultat
34     System.out.println();
35     System.out.println("La somme des " + nb + "er
36
37
38 }
39
```
- Bottom Panel:**
 - Variables: Shows the state of variables.

Name	Type	Value
<Enter new watch>		
Static		
args	String[]	#35(length=0)
nb	int	5
result	int	0
 - Output: Empty.
- Status Bar:** Exo2 (debug) running... 28:1 INS Windows (CRLF)

d) Testing the main code

The screenshot displays an IDE interface with the following components:

- Project Explorer (Left):** Shows a project named 'Exo2' with sub-packages 'Source Packages' and 'Libraries'. The 'Exo2.java' file is selected under 'Source Packages'.
- Source Editor (Center):** Displays the code for 'Exo2.java'. The code calculates the sum of the first 5 integers. The line `result=result+ind;` is highlighted in yellow.
- Debugger Console (Bottom):** Shows the execution output. The first line is 'run:'. The second line is 'La somme des 5entiers est: 15'. The third line is 'BUILD SUCCESSFUL (total time: 0 seconds)'.

```
14
15 /**
16  * @param args the command line arguments
17  */
18 public static void main(String[] args) {
19     // TODO code application logic here
20     //Declaration des variables
21     int nb; // nombre d'entier a additionner
22     int result; // resultat
23     int ind; //indice
24     nb=5;
25     result=0;
26
27     // Addition des nb premiers entiers
28     ind=1;
29     while (ind <= nb) {
30         result=result+ind;
31         ind++;
32     }
33
34     // Affichage du resultat
35     System.out.println();
36     System.out.println("La somme des "+ nb + " e
37 }
38
39
40
```

run:

La somme des 5entiers est: 15

BUILD SUCCESSFUL (total time: 0 seconds)

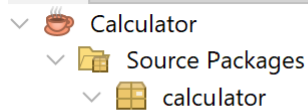
6. Testing the code

The screenshot displays an IDE interface with the following components:

- Top Menu Bar:** File, Edit, View, Navigate, Source, Refactor, Run, Debug, Profile, Team, Tools, W, Exo2...
- Toolbar:** Includes icons for file operations, running, and debugging.
- Left Panel (Project Explorer):** Shows a project structure with packages 'Exo2' and 'TP0'. The 'Exo2' package contains 'Exo2.java'. The 'TP0' package contains 'TP0.java'.
- Bottom Left Panel (Main - Navigator):** Shows the 'main' method of the 'Exo2' class, which takes 'String[] args' as input.
- Source Editor:** Displays the code for 'Exo2.java'. The code is as follows:

```
14  */
15  public class Exo2 {
16
17      /**
18       * @param args the command line arguments
19       */
20      public static void main(String[] args) {
21          // TODO code application logic here
22          //Declaration des variables
23          int nb; // nombre d'entier a additionner
24          //nb=5;
25          Scanner sc = new Scanner(System.in);
26          System.out.println("\n Enter le nombre :");
27          nb=sc.nextInt(); // On demande a sc de donner
28          int result; // resultat
29          int ind; //indice
30          result=0;
31
32          // Addition des nb premiers entiers
33          ind=1;
34          while (ind <= nb) {
35              result=result+ind;
36              ind++;
37          }
38
39          // Affichage du resultat
40          System.out.println();
```
- Bottom Right Panel (Output):** Shows the output of the program. It includes a 'run:' label, the input '4', the output 'La somme des 4 entiers est: 10', and a status message 'BUILD SUCCESSFUL (total time: 4 seconds)'.
- Bottom Status Bar:** Shows 'Finished building Exo2 (run)', '25:26', and 'INS'.

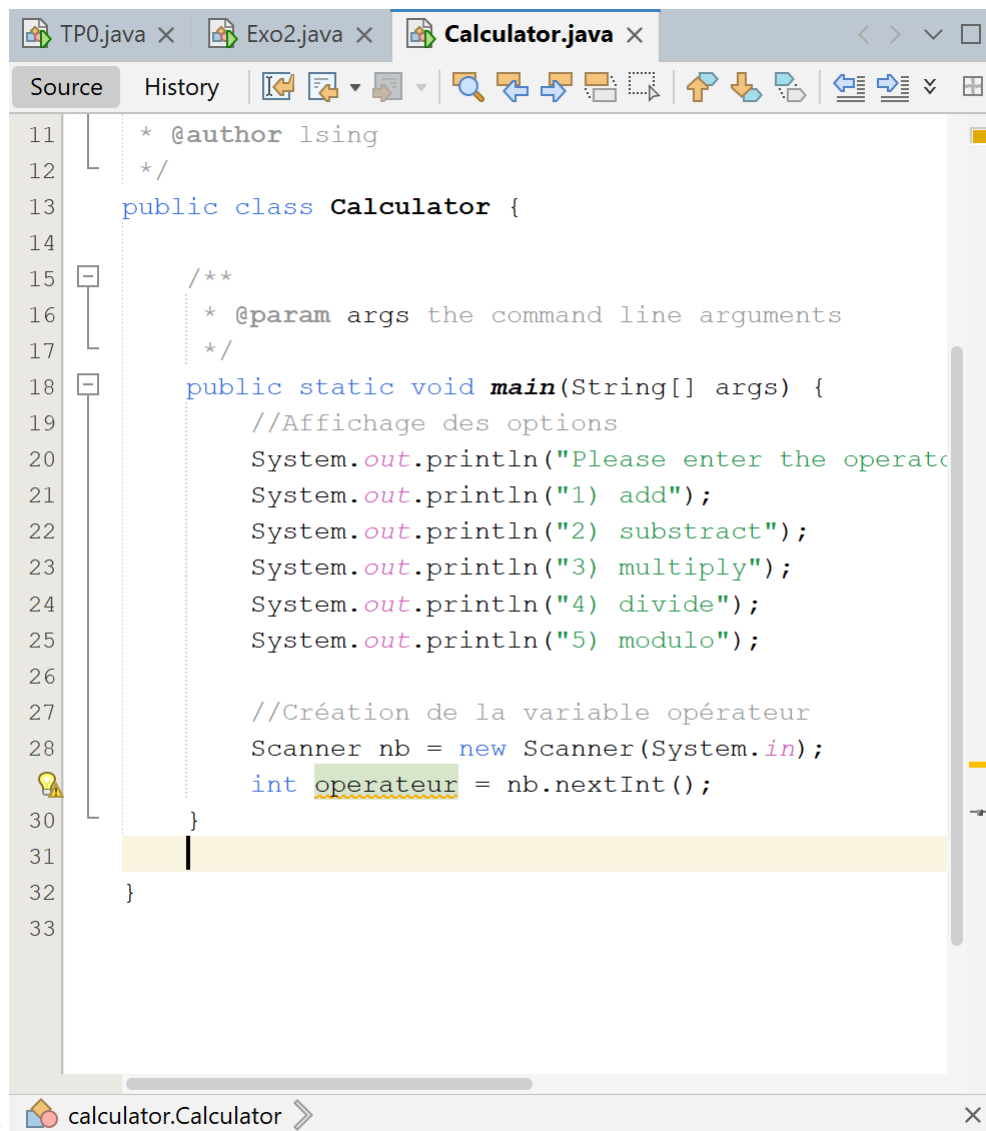
Partie 3 :



Ex1:  Calculator.java Creating the Calculator project

Ex2: Showing to the user different options

A screenshot of an IDE interface. The top menu bar includes File, Edit, View, Navigate, Source, Refactor, Run, Debug, Profile, Team, Tools, and W. Below the menu is a toolbar with various icons. The left sidebar shows the project explorer with a tree structure: Calculator (Source Packages) -> calculator -> Calculator.java. Below this is the 'main - Navigator' panel showing 'Members' for the 'Calculator' class, including 'Calculator()' and 'main(String[] args)'. The main editor area displays the source code of 'Calculator.java'. The code starts with 'package calculator;' and a class definition 'public class Calculator {'. Inside the class, there is a 'main' method that prints a menu of options: '1) add', '2) subtract', '3) multiply', '4) divide', and '5) modulo'. The bottom panel shows the 'Output' window with the text 'run: Please enter the operator' followed by the menu options. The status bar at the bottom indicates 'Finished building Calculator (run).', '23:41', and 'INS Windows (CRLF)'.



```
11  * @author lsing
12  */
13  public class Calculator {
14
15      /**
16       * @param args the command line arguments
17       */
18      public static void main(String[] args) {
19          //Affichage des options
20          System.out.println("Please enter the operator");
21          System.out.println("1) add");
22          System.out.println("2) subtract");
23          System.out.println("3) multiply");
24          System.out.println("4) divide");
25          System.out.println("5) modulo");
26
27          //Création de la variable opérateur
28          Scanner nb = new Scanner(System.in);
29          int opérateur = nb.nextInt();
30      }
31  }
32
33
```

Ex 3: calculator.Calculator > Creation of the operatuer value.

```
System.out.println("Choose a number");
Scanner nb1 = new Scanner(System.in);
int operand1 = nb1.nextInt();

System.out.println("Choose an another number");
Scanner nb2 = new Scanner(System.in);
int operand2 = nb2.nextInt();
```

Ex 4 + Ex 5: Creation of the operand1 and 2 values.

```
3) multiply
4) divide
5) modulo
3
Choose a number:
2
Choose an another number:
4
Result is :8
BUILD SUCCESSFUL (total time: 8 seconds)
```

Ex 6:

```
switch(operateur) {
    case '1':
        System.out.println("Result is :" + (operande1 + operande2));
        break;
    case '2':
        System.out.println("Result is :" + (operande1 - operande2));
        break;
    case '3':
        System.out.println("Result is :" + (operande1 * operande2));
        break;
    case '4':
        System.out.println("Result is :" + (operande1 / operande2));
        break;
    case '5':
        System.out.println("Result is :" + (operande1 % operande2));
        break;
}
```

```
" + ((float)operande1 / operande2));
```

the division

Add the switch case. And the float type for

Ex 7 + Ex 8 : Implementing the error message by if condition

```
if (operateur < 1 || operateur > 5) {
    System.out.println("Error : enter a number between 1 and 5");
    return;
}
```

```
1. 0.11.
Please enter the operator :
1) add
2) subtract
3) multiply
4) divide
5) modulo
6
Error : enter a number between 1 and 5
BUILD SUCCESSFUL (total time: 3 seconds)
```

Example of the error message